

Kiosk Vision

An AI-powered accessible kiosk assistant | Feature Build Sequence & Team Plan

Self-service kiosks are widely used in restaurants, retail, and pharmacies, but often fail specially-abled users. Kiosk Vision builds an AI-powered assistant that makes these kiosks inclusive through voice, visual, and adaptive interaction — while also solving practical bottlenecks like payment handling.

This document lists the 10 core features in build order, followed by a role-based split across a team of 5.

Feature Build Sequence

Features are ordered so that each phase produces a usable, demoable system before moving to the next layer of complexity.

Phase 1 — Core Working Kiosk (Weeks 1–4)

1. Conversational Voice Ordering (Core)

Full voice-driven menu navigation and ordering. Speech-to-text captures what the user says, an LLM parses the intent into a structured order action, and text-to-speech reads back the order and total before payment. This is the backbone feature every visually-impaired user depends on, so it must be built first.

2. Real-Time Screen Understanding

A vision-language model interprets whatever is currently on screen and narrates it in plain language, so the assistant keeps working even as the menu or UI changes. This is the most technically distinctive feature in the project.

3. Simplified / Adaptive UI Mode

Detects confusion or low digital familiarity (long idle time, repeated back-and-forth) and switches to a step-by-step, icon-first, one-choice-at-a-time layout.

10. Graceful Human Handoff

When AI confidence is low or the user is clearly struggling, the system cleanly escalates to a human attendant instead of trapping the user in a broken loop. Built early so every other feature has a safety net to fall back on.

Phase 2 — Practical Experience Layer (Weeks 5–8)

6. Assistive Payment Guidance

Step-by-step audio and visual guidance through payment (“insert your card now”, “tap now”), reading the amount aloud before confirming the charge. The real payment gateway is simulated; the focus is the guided user experience.

7. Automatic Need Detection

Detects signs that a user needs help — hesitation, no interaction, repeated failed taps, or optionally a cane/wheelchair via computer vision — and proactively offers the right mode. Always paired with a simple manual “I need help” override.

9. Real-Time Captioning Display

Live captions of everything the kiosk says out loud, plus captions for detected ambient announcements. Directly serves hearing-impaired users and reuses the speech pipeline already built in Phase 1.

Phase 3 — Advanced Differentiators (Weeks 9+, pick 1–2 to go deep on)

4. AVSR / Lip-Reading-Assisted Voice Recognition

Uses MediaPipe Face Mesh to track lip movement and fuses it with audio, keeping voice commands reliable in noisy environments (busy restaurants, stations, airports).

5. Gaze Tracking + Dwell Selection

Webcam-based gaze estimation lets users “click” by staring at a button for 1–2 seconds. Ship with a short visible calibration step rather than claiming true zero-calibration.

8. Multilingual + Accent-Robust Voice

Support for regional languages and Hindi-English code-mixing, tested specifically against noisy, real-store conditions for reliability.

Note: rPPG-based stress detection and depth-sensor spatial adjustment were considered but intentionally excluded from the core plan — the former raises privacy/health-data concerns, the latter requires specialized depth-sensing hardware. Both can be mentioned as future scope.

Team Split (5 People)

Roles are grouped by skill area rather than splitting the 10 features evenly, since several features share the same underlying pipeline.

Person 1 — Voice & Conversational AI Lead

Owns: Feature 1, Feature 8

- Speech-to-text integration (e.g. Whisper or similar)
- LLM-based intent parsing (spoken request → structured order action)
- Text-to-speech responses and order confirmation
- Regional language and Hindi-English code-mixed support

Person 2 — Computer Vision Lead

Owns: Feature 4, Feature 7

- MediaPipe Face Mesh setup for lip movement tracking
- Fusing lip movement signal with audio confidence scoring
- Idle-time / hesitation detection logic
- Optional: cane / wheelchair detection via object detection

Person 3 — Screen Understanding & Vision-Language Lead

Owns: Feature 2, Feature 5

- Vision-language pipeline to read and describe current screen state
- Screenshot capture and narration logic
- Gaze vector estimation and dwell-time click implementation
- Most research-heavy role — best paired with strong ML interest

Person 4 — Frontend / UI Lead

Owns: Feature 3, Kiosk UI shell

- Build the kiosk interface (React) that every other module plugs into
- Implement adaptive/simplified UI states (icons, large text, step-by-step flow)
- Wire up voice, gaze, and CV outputs to trigger real UI actions
- Acts as the integration hub — coordinates closely with all other roles

Person 5 — Backend, Payments & Accessibility Output Lead

Owns: Feature 6, Feature 9, Feature 10

- Backend API for order state, menu data, and sessions
- Simulated payment flow with audio confirmation step
- Live captioning system, syncing STT output to an on-screen display
- Confidence-threshold logic for escalating to human help

Coordination tip: Person 4 (Frontend) is the integration hub — voice, CV, screen understanding, and backend all feed into or read from the kiosk UI. Get Person 4 and Person 5 building the basic kiosk and backend skeleton in Week 1 so Persons 1, 2, and 3 have a real system to plug their AI modules into rather than working in isolation.